

IBRAHIM ALI GHANIM

MECHATRONICS ENGINEER | OIL & GAS AUTOMATION, SCADA & INSTRUMENTATION

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (2026), ranked 9th out of 90 students (Top 10%), focused on oil and gas automation, SCADA, instrumentation, and electrical control. Project experience in tank and pump monitoring, industrial alarms, Modbus TCP, Siemens PLC/HMI, Node-RED, Factory I/O, and AI-assisted anomaly detection. Practical exposure to SCADA control rooms, electrical panels, instrument calibration, and troubleshooting.

EDUCATION

B.Sc. in Mechatronics Engineering | University of Baghdad - Al-Khwarizmi College of Engineering | June 2026
Ranked 9th out of 90 Mechatronics Engineering students (Top 10% of the cohort)

TECHNICAL SKILLS

PLC & Industrial Automation: Siemens S7-1200/S7-200, Schneider Electric Zelio, TIA Portal, PLCSIM, Ladder Logic, Classic Control, Sequential Control, Interlocks, PID Fundamentals, E-STOP, Process Trip

SCADA & HMI: Ignition 8.3, Siemens WinCC, Node-RED, Factory I/O, OPC UA, Tags, Alarming, Tag Historian, Trends, Event Logging, Manual/Auto Control, Vision & Perspective Fundamentals

Instrumentation & Process Control: Level/Flow/Pressure Instrumentation, 4-20 mA, 0-10 V, Calibration, Control Valves, Pumps, Instrument Air, Pneumatic Systems, Leak/Blockage Detection, P&ID/PFD Interpretation

Industrial Communication: Modbus TCP, OPC UA, Ethernet/IP, PLC-HMI Communication, REST/JSON, Industrial Data Integration

Electrical & Motor Control: Electrical & Control Schematic Reading, Relays, Contactors, Circuit Breakers, Motor Starters, Control Panels, VFD/Soft Starter Fundamentals

AI for Process Monitoring: Python, FastAPI, Isolation Forest, Anomaly & Fault Detection

CERTIFICATIONS

[Ignition 8.3 Credential](#) | Inductive University | 2026

[One Million Prompts](#) | Dubai Future Foundation | 2026

INTERNSHIPS & TECHNICAL EXPERIENCE

SCADA & Electrical Systems Intern | Ministry of Electricity - Rusafa Directorate, Baghdad | Jul 2025 - Aug 2025

Observed SCADA control-room monitoring, alarm response, server infrastructure, electrical control panels, and the connection between field equipment and centralized supervision.

Instrumentation & Maintenance Trainee | Alchemy Scientific, Baghdad | Jun 2026 - Jul 2026

Performed calibration, functional testing, basic troubleshooting, probe care, and battery replacement on pH, ORP, EC, TDS, temperature, turbidity, and photometer instruments.

SELECTED PROJECTS

PetroSafe AI - Industrial Tank Safety & Anomaly Detection | Independent Project | [View Project](#)

Built an industrial tank safety prototype using Factory I/O, Node-RED, Modbus TCP, FastAPI, and Isolation Forest, with Auto/Manual control, E-STOP, Process Trip, fault latching, leak/blockage detection, alarms, trends, and incident logging.

Instrument Air & CHU Monitoring Dashboard | Oil & Gas Case Study | [View Project](#)

Developed a Node-RED dashboard for Instrument Air and CHU monitoring, covering pressure, dew point, airflow, particulate contamination, compressor/dryer status, alarms, and desiccant health.

Multi-Station Tank & Pump Control System | Industrial Process Simulation | [View Project](#)

Simulated 16 tanks and four pumping substations using Siemens PLC/HMI, TIA Portal, PLCSIM, Factory I/O, Node-RED, Modbus, automatic pump logic, alarms, trends, leak detection, and P&ID/PFD interpretation.

LANGUAGES

Arabic - Native | English - Intermediate